

Nikhil Tiwari .Net Full Stack Developer

📍 Bangalore ✉ nikhiltiwari1057@gmail.com ☎ 8266929986 🌐 nikhiltiwari1057

PROFILE

Full Stack Developer with 7+ years building healthcare integration platforms, cloud microservices, and production-grade web applications. Architected multi-service systems from zero on Azure and AWS. spanning .NET backends, Python APIs, Vue/Angular frontends, and cloud infrastructure. Deep expertise in healthcare data workflows with a track record of designing resilient, cache-optimized systems at scale.

SKILLS

Languages: C# | Python | TypeScript/JavaScript | SQL | HTML/CSS

Backend Technical: .NET 8 | Core Web API | FastAPI | Entity Framework | LINQ | Middleware Design | SOAP↔REST Conversion | X12/EDI Parsing

Frontend Technical: Angular | JavaScript | HTML | CSS | Vue 3 + Vite

Cloud and Tools: Azure DevOps | Azure Cloud Services | ECS/EKS | Lambda | Terraform | Kubernetes | Docker

Databases and Devops: Azure Cosmos DB, SQL Server, Databricks/PySpark | CI/CD (Azure Pipelines, GitHub Actions, GitLab CI) | Polly, TYK Gateway, IIS, Git

AI: ONNX Runtime | tree-sitter | MCP Protocol | Semantic Search

Domain: Healthcare | Insurance | Leasing | Loan

PROFESSIONAL EXPERIENCE

Availity L.L.C.

2025/07 – Present | Bangalore

Software Developer

- **Architected a multi-service healthcare platform from zero (Mock Payer Platform)** - 5 microservices, Vue 3 UI, Azure Functions, CI/CD pipelines
- Implemented cache-first provider search with Cosmos DB and Polly retry with exponential backoff, reducing payer API calls and improving response times
- Built X12/EDI eligibility parsing with payer-specific strategy pattern - extensible architecture enabling new payer onboarding without modifying existing code
- **Architected cross-cloud reverse proxy on AWS EKS routing to Azure backends (TYK/APIM)**, with dual-lane routing, SOAP↔REST conversion, health-check failover (10s TTL), and PHI-safe in-memory EDI handling
- **Reduced document search latency via dual-layer caching (TTL-based, stable JSON hashing)** and designed 7 covering indexes eliminating full table scans on a FastAPI/Databricks service
- **Deployed and managed full Azure infrastructure** - App Service, APIM, Cosmos DB, Functions, Key Vault, Entra ID -plus CI/CD with PR-triggered builds and per-service test gates

Sapiens

2022/09 – 2025/07 | Bangalore

Senior Developer

- Lead development and maintenance of **Underwriting Pro**, focusing on both **client support** and **R&D initiatives**.
- Integrated **OKTA SSO** for seamless and secure authentication flow using OAuth 2.0 and JWT tokens.
- Implemented role-based access control (RBAC) leveraging OKTA groups and claims mapping.
- Collaborated with the security and compliance teams at Farmers Insurance to ensure OKTA configurations aligned with enterprise security standards.
- Continue as the **main point of contact** for **ERIE Insurance**, implementing advanced features and ensuring seamless system functionality.
- Drive upgrade projects for clients like **Federated**, **Erie**, and **AAFMAA**, enhancing system stability and performance.
- Contribute to ongoing **Angular upgrades**, improving UI/UX and application efficiency.

Odessa inc.

2019/04 – 2022/09

Senior Software Engineer

- Developed API systems and backend solutions using **C#, Azure, SQL, .NET Framework, .NET Core, ASP.NET** , and **LINQ**.
- Designed and implemented features like **Asset Split** by converting SQL scripts into C# classes.
- Built solutions such as **Auto-Payoff**, **Offline Background Processes**, and **SKU extensions** for major clients like **Dell** and **HP**.
- Created **SQL scripts** for the **reconciliation of GL journals** and optimized data processes.
- Managed environments using **IIS Server** for deployment and maintenance.
- Led **regression testing**, developed **automation test cases**, and mentored interns to bridge technical gaps.

PROJECTS

MNEMO — Persistent Memory for AI Coding Agents

Creator & Sole Developer | Published on PyPI, NPM, Brew, VS CODE extention

- Open-source tool giving AI coding agents (Claude, Kiro, Cursor, Copilot)
- persistent memory, knowledge graphs, and semantic search across sessions.

- Built complete developer tool from concept to published package — 14,000+ lines
- of Python, published to PyPI (pip install mnemo-dev), VS Code extension on marketplace
- Achieved 100% search recall@5 with 2ms latency — triple-stream search fusing
- ONNX dense embeddings, BM25 keyword search, and Dijkstra graph traversal
- Engineered 8x init speedup — optimized tree-sitter AST parsing pipeline
- (14 languages) from ~56s to ~7s for a 300-file repo
- Built knowledge graph with community detection (NetworkX + Leiden clustering)
- Implemented lifecycle hooks for 5 AI clients with client-specific configuration
- Maintained 222 passing tests covering search quality, memory lifecycle, and MCP integration

Tech: Python, ONNX Runtime, tree-sitter, NetworkX, MCP Protocol, Next.js, TypeScript

KIRO HARNESS ENGINEERING

Replicated multi-agent harness architecture for autonomous website generation using Generator↔Evaluator adversarial loop.

- Built harness orchestrator spawning fresh Kiro CLI sessions per agent with
- file-based inter-agent communication
- Integrated Playwright MCP for live headless site testing and scoring
- Produced a complete marketing website through 12 adversarial iterations in 3.5 hours with zero manual coding

Tech: Shell, Kiro CLI, Playwright MCP, Next.js, Tailwind CSS, Framer Motion

EDUCATION

B.tech (Computer Science Engineering) GLA University CGPA: 8.0/10	2015 – 2019
XII (SENIOR SECONDARY) Holy Public School CBSE Board Percentage: 82.40%	2012 – 2013
X(SECONDARY) Holy Public School CBSE Board GPA: 9.20/10	2010 – 2011

CERTIFICATES

- AZ- 900